

AP-E16 Tangent Defects, Purified Caccioppoli Control, and a Homogeneous Hopf-Base Young-Measure Obstruction

Route-F derivation ledger

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Abstract

This note executes the AP-E15 continuation without reopening the lattice branch. The complete-minor relaxation defect is split into five nonnegative Hopf Radon measures. At μ -almost every point, one normalized tangent sequence carries all five components with their Radon–Nikodym weights, and a diagonal smooth recovery sequence can be chosen so that its normalized compact-fibre-phase relaxation drop tends to zero.

A new scaling obstruction appears. If the spatial radius is r , the target amplitude is s , and first-, second-, and third-minor energies all contribute to one defect mass, then their local scales are

$$s^2 r, \quad s^4 / r, \quad s^6 / r^3.$$

They admit one ordinary field blow-up only when $s \sim r$ and $\mu(B_r) \sim r^3$. Nonvolumetric tangents must be treated as order-specific measure or generalized-Young-measure limits.

Inside one normal target ball, a degree-preserving geodesic cutoff gives a conditional Caccioppoli inequality with exactly the AP-E13 weighted terms and the normalized local comparison deficit. It does not by itself yield excess contraction. Indeed an explicit smooth sphere-valued family, after exact fibre-phase purification, generates the homogeneous pure-base Young measure

$$F_{31} = \cos \theta_1, \quad F_{42} = \cos \theta_2.$$

Its barycentric gradient and minors vanish, its weighted cutoff tends to zero, but its diffuse graph-energy defect is $\frac{1}{2} + \frac{R}{8} > 0$. This disproves any unconditional inference “phase purification plus two-limit weighted decay implies $\mu = 0$ ”. The family is not asymptotically minimizing, so it does not refute the desired theorem for the selected $B = 1$ minimizer. It proves that normalized full local quasiminimality and a strong-quasiconvex boundary comparison are indispensable. Hessian, determinant, and portal promotion remain closed.

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1 Frozen energy and the Hopf defect vector

The AP-E11 action remains

$$\mathcal{E}(u) = \int_{\Omega} \left[\frac{1}{2} |Du|^2 + \frac{R}{2} |M_2(Du)|^2 + \frac{K}{2} |M_3(Du)|^2 + m^2 (1 - u^0) \right] dx, \\ (R, K, m^2) = (1, 0.35, 0.12). \quad (1)$$

Let u_j be a smooth boundary-preserving degree- B sequence with $\mathcal{E}(u_j) \rightarrow I_B$, strongly convergent in L^2 to a graph representative u , and weakly convergent at all three complete-minor levels. AP-E15 defines

$$e_j dx \xrightarrow{*} e_u dx + \mu, \quad e_j = \frac{1}{2} |Du_j|^2 + \frac{R}{2} |M_2(Du_j)|^2 + \frac{K}{2} |M_3(Du_j)|^2, \quad \mu \geq 0. \quad (2)$$

For $n_j = h(u_j)$ and $a_j = 2u_j^* \theta$, set

$$\begin{aligned} \nu_j^a &= \frac{1}{8} |a_j|^2 dx, & \nu_j^n &= \frac{1}{8} |Dn_j|^2 dx, \\ \nu_j^{an} &= \frac{R}{32} |a_j \wedge Dn_j|^2 dx, & \nu_j^c &= \frac{R}{32} |da_j|^2 dx, \\ \nu_j^{cs} &= \frac{K}{128} |a_j \wedge da_j|^2 dx. \end{aligned} \quad (3)$$

The exact AP-E14 identities imply

$$e_j dx = \nu_j^a + \nu_j^n + \nu_j^{an} + \nu_j^c + \nu_j^{cs}. \quad (4)$$

Strong convergence of u_j and weak graph convergence imply weak L^2 convergence of each Hopf factor to its counterpart for u . Weighted lower semicontinuity therefore gives, after one subsequence,

$$\nu_j^\beta \xrightarrow{*} \nu_u^\beta + \mu_\beta, \quad \mu_\beta \geq 0, \quad \mu = \sum_{\beta \in \{a, n, an, c, cs\}} \mu_\beta. \quad (5)$$

Definition 1.1 (Hopf Radon–Nikodym weights). *Because every $\mu_\beta \leq \mu$, define*

$$\vartheta_\beta(x) = \frac{d\mu_\beta}{d\mu}(x), \quad \vartheta_\beta \geq 0, \quad \sum_\beta \vartheta_\beta = 1 \quad \mu\text{-a.e.} \quad (6)$$

These weights distinguish fibre, base, shear, curvature, and Chern–Simons-square defect. Fibre-phase purification does not assert $\vartheta_a = \vartheta_{an} = \vartheta_{cs} = 0$; it removes only the part exposed by an exact compact phase relaxation.

2 A common μ -a.e. tangent and diagonal recovery

Let $T_{x_0, r}(x) = (x - x_0)/r$. At μ -almost every x_0 , standard doubling-radius selection gives $r_\ell \downarrow 0$ such that the normalized measures

$$\mu_{x_0, r_\ell} = \frac{(T_{x_0, r_\ell})_\# \mu}{\mu(B_{r_\ell}(x_0))} \xrightarrow{*} \tau \neq 0 \quad \text{locally in } \mathbb{R}^3. \quad (7)$$

The normalization can be chosen with $\tau(B_1) = 1$ after avoiding boundary-mass radii.

Theorem 2.1 (Common Hopf tangent and phase-stationary diagonal). *For μ -almost every x_0 , the radii in (7) may be chosen so that*

$$\frac{(T_{x_0, r_\ell})_\# \mu_\beta}{\mu(B_{r_\ell}(x_0))} \xrightarrow{*} \vartheta_\beta(x_0) \tau \quad \text{for every } \beta. \quad (8)$$

There is also a diagonal $j_\ell \rightarrow \infty$ for which the scaled signed recovery defects converge to the same measures and, for every fixed $L > 0$,

$$\frac{\Delta_{j_\ell}(B_{Lr_\ell}(x_0))}{\mu(B_{r_\ell}(x_0))} \longrightarrow 0. \quad (9)$$

Thus every tangent is stationary, in normalized energy, against compact exact fibre shifts.

Proof. The Radon–Nikodym differentiation theorem for the finite family $\mu_\beta = \vartheta_\beta \mu$ gives

$$\frac{\mu_\beta(B_r(x_0))}{\mu(B_r(x_0))} \longrightarrow \vartheta_\beta(x_0) \quad \text{for } \mu\text{-a.e. } x_0.$$

Apply this simultaneously to a countable dense set of compactly supported test functions after the doubling-radius selection. This proves (8).

Let $\sigma_j^\beta = \nu_j^\beta - \nu_u^\beta$. For each already chosen r_ℓ , weak-star convergence permits j_ℓ to be so large that the scaled errors against the first ℓ test functions are smaller than $1/\ell$. Since $\mathcal{E}(u_j) - I_B \rightarrow 0$, enlarge j_ℓ again so that

$$\frac{\mathcal{E}(u_{j_\ell}) - I_B}{\mu(B_{r_\ell}(x_0))} < \frac{1}{\ell}. \quad (10)$$

AP-E15 proves $0 \leq \Delta_j(Q) \leq \mathcal{E}(u_j) - I_B$ on every fixed admissible phase ball. For fixed L , Lr_ℓ eventually lies below the phase-convexity radius. Equations (10) and the AP-E15 bound give (9). $\square$

Remark 2.2 (Precise consequence of purification). *Equation (9) is stronger than unnormalized stationarity, but weaker than strong convergence of a_j . A co-closed or base-induced connection oscillation can be stationary along the exact fibre orbit. The tangent weights in (6) must therefore remain in the ledger until a full local comparison eliminates them.*

3 Why one field blow-up exists only in the volumetric branch

Suppose a tangent is represented in a target chart by

$$u(x_0 + ry) = \exp_q(s v(y)), \quad (11)$$

where r is the spatial scale and s the target amplitude. Up to uniformly bounded chart factors, homogeneity of complete minors gives

$$\begin{aligned} \int_{B_r} |Du|^2 dx &\sim s^2 r \int_{B_1} |Dv|^2 dy, \\ \int_{B_r} |M_2(Du)|^2 dx &\sim \frac{s^4}{r} \int_{B_1} |M_2(Dv)|^2 dy, \\ \int_{B_r} |M_3(Du)|^2 dx &\sim \frac{s^6}{r^3} \int_{B_1} |M_3(Dv)|^2 dy. \end{aligned} \quad (12)$$

Proposition 3.1 (Simultaneous-normalization obstruction). *Assume $s = r^\alpha$ and $\mu(B_r) \asymp r^d$. All three expressions in (12) can be simultaneously of order $\mu(B_r)$ only if*

$$2\alpha + 1 = d, \quad 4\alpha - 1 = d, \quad 6\alpha - 3 = d, \quad (13)$$

whose unique solution is

$$\boxed{\alpha = 1, \quad d = 3.} \quad (14)$$

Proof. The exponents in (13) are the powers of r in (12). Subtracting the first equation from the second gives $2\alpha = 2$; hence $\alpha = 1$, and then $d = 3$. The third equation is consistent and gives no second branch. $\square$

Thus a volumetric defect can have an ordinary tangent-gradient Young measure with $s \sim r$. At a nonvolumetric point, normalizing the first-order field necessarily sends at least one higher-minor scale to zero or infinity. The correct object is then the component tangent (8), or a generalized Young measure with separate concentration coordinates. Treating every tangent as one $W^{1,2}$ field would silently discard precisely the endpoint concentration under study.

4 Conditional purified Hopf-base Caccioppoli inequality

For a ball $B_t \Subset \Omega$, define the relative-homotopy local comparison deficit

$$\epsilon_j(B_t) = \mathcal{E}(u_j; B_t) - \inf \left\{ \mathcal{E}(v; B_t) : \begin{array}{l} v = u_j \text{ near } \partial B_t, \\ v \simeq u_j \text{ relative } \partial B_t \end{array} \right\}. \quad (15)$$

Replacing u_j by an allowed v in the full map preserves boundary data and global degree. Consequently

$$0 \leq \epsilon_j(B_t) \leq \mathcal{E}(u_j) - I_B. \quad (16)$$

Let the image of u_j on B_t lie in one fixed normal ball centred at $q \in \mathbb{S}^3$, and put $\xi_j = \log_q u_j$. For $0 < s < t$, choose $\eta = 1$ on B_s , $\eta = 0$ outside B_t , and $|D\eta| \leq 2/(t-s)$. The comparison

$$v_j = \exp_q((1-\eta)\xi_j) \quad (17)$$

equals q in B_s , equals u_j near ∂B_t , and is homotopic to u_j relative to the boundary.

Lemma 4.1 (Rank-one cutoff minor bounds). *With $A = D\xi_j$, $B = -\xi_j \otimes D\eta$, and $c = 1 - \eta$,*

$$\begin{aligned} |D((1-\eta)\xi_j)|^2 &\leq C(c^2|A|^2 + |\xi_j|^2|D\eta|^2), \\ |M_2(cA + B)|^2 &\leq C(c^4|M_2(A)|^2 + c^2|\xi_j|^2|D\eta|^2|A|^2), \\ |M_3(cA + B)|^2 &\leq C(c^6|M_3(A)|^2 + c^4|\xi_j|^2|D\eta|^2|M_2(A)|^2). \end{aligned} \quad (18)$$

Proof. B has rank one. Hence

$$M_2(cA + B) = c^2 M_2(A) + c L_2(A, B), \quad M_3(cA + B) = c^3 M_3(A) + c^2 L_3(A, A, B),$$

because every term containing two columns from B vanishes. Exterior algebra gives $|L_2|^2 \leq C|A|^2|B|^2$ and $|L_3|^2 \leq C|M_2(A)|^2|B|^2$. Together with $|B| = |\xi_j||D\eta|$, this proves (18). The derivatives of $\exp_q$ and its inverse only change C on the fixed normal ball. $\square$

Write

$$\Phi_j(r) = \int_{B_r} \left[\frac{1}{2} |Du_j|^2 + \frac{R}{2} |M_2(Du_j)|^2 + \frac{K}{2} |M_3(Du_j)|^2 \right] dx. \quad (19)$$

Theorem 4.2 (Conditional Caccioppoli/hole-filling estimate). *There is a constant C depending only on the normal ball and the frozen coefficients such that*

$$\begin{aligned} \Phi_j(s) &\leq C [\Phi_j(t) - \Phi_j(s)] \\ &\quad + \frac{C}{(t-s)^2} \int_{B_t \setminus B_s} |\xi_j|^2 (1 + R|Du_j|^2 + K|M_2(Du_j)|^2) dx \\ &\quad + Cm^2 \int_{B_t} |\xi_j| dx + \epsilon_j(B_t). \end{aligned} \quad (20)$$

Equivalently, with $\theta = C/(1+C) < 1$,

$$\begin{aligned} \Phi_j(s) &\leq \theta \Phi_j(t) + \frac{C}{(t-s)^2} \int_{B_t \setminus B_s} |\xi_j|^2 (1 + R|Du_j|^2 + K|M_2(Du_j)|^2) dx \\ &\quad + Cm^2 \int_{B_t} |\xi_j| dx + C\epsilon_j(B_t). \end{aligned} \quad (21)$$

Proof. Use (17) in (15). The derivative energy of v_j vanishes on B_s . Cancel the unchanged exterior, discard the nonnegative derivative energy of u_j on the annulus, and apply (18). Because $V(u) = m^2(1 - u^0)$ is m^2 -Lipschitz on $\mathbb{S}^3$,

$$\int_{B_t} |V(v_j) - V(u_j)| dx \leq Cm^2 \int_{B_t} |\xi_j| dx.$$

This proves (20). Moving $C\Phi_j(s)$ to the left gives (21). $\square$

After AP-E15 purification, the exact fibre-phase part of the normalized comparison deficit vanishes. The estimate above is therefore the legitimate local inequality for the remaining base/mixed problem. It still needs four inputs before it becomes an excess theorem:

1. normal-ball control or a trace-preserving endpoint replacement;
2. sequence-uniform control of the weighted second- and third-order cutoff terms;
3. normalized full local quasiminimality, not only phase stationarity;
4. a strong-quasiconvex harmonic-approximation step which improves the scale exponent, rather than only $\theta < 1$ hole filling.

5 A homogeneous phase-purified base Young measure

The preceding list is not a technical luxury. There is an exact obstruction if the third input is omitted. On $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$, for an integer $N \geq 2$, define

$$u_N(x) = \left(\sqrt{1 - \frac{\sin^2(Nx_1) + \sin^2(Nx_2)}{N^2}}, 0, \frac{\sin(Nx_1)}{N}, \frac{\sin(Nx_2)}{N} \right). \quad (22)$$

It lies exactly in $\mathbb{S}^3$, converges uniformly to e_0 , has rank at most two, and is null-homotopic because its image lies in the open hemisphere $u^0 > 0$.

Proposition 5.1 (Positive diffuse pure-base defect). *The family (22) satisfies*

$$\begin{aligned} Du_N &\rightharpoonup 0, & M_2(Du_N) &\rightharpoonup 0, & M_3(Du_N) &= 0, \\ \frac{1}{|\mathbb{T}^3|} \int_{\mathbb{T}^3} |Du_N|^2 dx &\longrightarrow 1, & \frac{1}{|\mathbb{T}^3|} \int_{\mathbb{T}^3} |M_2(Du_N)|^2 dx &\longrightarrow \frac{1}{4}. \end{aligned} \quad (23)$$

Its graph-energy measures converge to the nonzero diffuse defect

$$e_N dx \xrightarrow{*} \left(\frac{1}{2} + \frac{R}{8} \right) dx. \quad (24)$$

Meanwhile its Hopf connection is

$$a_N = \frac{2}{N} [-\sin(Nx_2) \cos(Nx_1) dx_1 + \sin(Nx_1) \cos(Nx_2) dx_2], \quad (25)$$

so $\|a_N\|_2 = O(N^{-1})$, while

$$da_N = 4 \cos(Nx_1) \cos(Nx_2) dx_1 \wedge dx_2. \quad (26)$$

The leading defect is therefore Hopf-base derivative/curvature, not an exact fibre oscillation.

Proof. Write $\theta_i = Nx_i$. The leading tangent derivatives are

$$\partial_1 u_N = (0, 0, \cos \theta_1, 0) + O(N^{-1}), \quad \partial_2 u_N = (0, 0, 0, \cos \theta_2) + O(N^{-1}).$$

All other columns vanish. Periodic averaging gives $\langle \cos \theta_i \rangle = 0$, $\langle \cos^2 \theta_i \rangle = 1/2$, and $\langle \cos^2 \theta_1 \cos^2 \theta_2 \rangle = 1/4$, proving (23)–(24). Direct substitution in $a = 2u^*\theta$ gives (25); differentiation gives (26). $\square$

Let χ_N minimize the periodic fibre-phase functional of u_N . Since $z_N = u_N^0 > 0$ is real, the potential contribution relative to $\chi = 0$ is $m^2 z_N (1 - \cos \chi) \geq 0$. Expanding every derivative term in $a_N + 2d\chi$ gives a coercive quadratic form in $d\chi$ plus a linear form bounded by $C\|a_N\|_2\|d\chi\|_2$. Comparison with $\chi = 0$ therefore yields

$$\|d\chi_N\|_2 \leq C\|a_N\|_2 = O(N^{-1}). \quad (27)$$

Elliptic regularity makes χ_N smooth. Thus $\tilde{u}_N = e^{i\chi_N} u_N$ is exactly phase-purified and has the same leading defect.

The associated homogeneous tangent-gradient Young measure is the pushforward of $(2\pi)^{-2} d\theta_1 d\theta_2$ by

$$F(\theta_1, \theta_2) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \cos \theta_1 & 0 & 0 \\ 0 & \cos \theta_2 & 0 \end{pmatrix}. \quad (28)$$

It obeys

$$\langle F \rangle = 0, \quad \langle M_2(F) \rangle = 0, \quad \left\langle \frac{1}{2}|F|^2 + \frac{R}{2}|M_2(F)|^2 \right\rangle = \frac{1}{2} + \frac{R}{8}. \quad (29)$$

It is therefore a nontrivial homogeneous complete-minor Young measure with zero barycentric graph vector.

Most importantly,

$$\frac{1}{|\mathbb{T}^3|} \int_{\mathbb{T}^3} |u_N - e_0|^2 |M_2(Du_N)|^2 dx = O(N^{-2}) \longrightarrow 0. \quad (30)$$

Hence even the AP-E15 two-limit weighted quantity can vanish while a diffuse base defect remains positive.

Theorem 5.2 (Exact scope of the no-go). *Phase purification, weak complete-minor graph convergence, and sequence-uniform weighted cutoff decay do not imply $\mu = 0$ or unconditional normalized excess contraction. The family $\tilde{u}_N$ is a precise counterexample.*

It is not a counterexample to the desired theorem for the selected minimizer: u_N has degree zero and positive energy while the constant map has zero energy. Equivalently, its normalized full local comparison deficit does not vanish. The construction may be inserted into a vacuum region of a smooth degree-one map without changing degree, but it still adds a removable positive energy.

6 Gate decision and next theorem

AP-E16 changes the blocker in three ways:

1. The μ -a.e. common tangent, Hopf component weights, and normalized phase-stationary diagonal are **proved**.
2. The geodesic-cutoff Caccioppoli inequality (20) is **proved conditionally** in one normal target ball, with every missing term displayed.
3. Unconditional excess contraction from phase purification and weighted cutoff decay is **disproved** by the homogeneous base Young measure (28).

The next theorem must combine, rather than separate, recovery and quasiminimality:

1. On the volumetric branch (14), prove a boundary-modification theorem which transfers normalized local quasiminimality to the tangent Young measure:

$$\frac{\epsilon_{j_\ell}(B_{r_\ell})}{\mu(B_{r_\ell})} \longrightarrow 0.$$

Exact strong quasiconvexity would then force that measure to be Dirac and eliminate the diffuse branch.

2. On nonvolumetric branches, work with the order-specific tangents (8); prove a concentration-capacity or topological lower-bound alternative rather than forcing one field normalization.
3. Only after both branches are excluded may one assert full $\mu = 0$, local recovery, regularity, and isolation.

The bosonic Hessian remains unauthorized. Dirac/Callias operators on a prescribed smooth Hopf base may continue as parallel mathematics, but determinant and degree-one portal promotion remain false.

Reproducibility. Run `python3 route_f/code/verify_ap_e16_tangent_caccioppoli_young_measure.py` from the repository root. The production card checks the scaling system, rank-one minor algebra, homogeneous Young measure, exact sphere generator, weighted-cutoff asymptotics, and all fail-closed gates. It performs no lattice relaxation or grid scan.